

Jackson Parrack

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Education

Georgia Institute of Technology | Atlanta, GA

Expected May 2029

Master of Science in Computer Science (part-time), Computing Systems

Saint Louis University | St. Louis, MO

Aug. 2022 - May 2026

Bachelor of Science in Computer Engineering, Minor in Computer Science

GPA: 3.93/4.0

Sogang University | Seoul, South Korea

Feb. 2024 - Jun. 2024

Exchange Program

Experience

Mastercard | O'Fallon, MO

Software Engineer I

Aug. 2026 - Present

- Building and scaling an internal MCP-based tool that gives 140+ employees natural language agentic access to company systems, letting teams query data directly rather than through engineers.
- Leading design and planning of a pilot transaction-insights agent targeting enterprise-wide adoption, combining LLM tool orchestration, MCP-based database access, input guardrails, and conversation memory.
- Developing a modernized clearing platform using Java Spring microservices on Kubernetes, replacing mainframe technology with services that scale horizontally to over 10M transactions/day.
- Maintaining and extending a Go-based simulation tool that tests the full payment lifecycle end-to-end, keeping it current with evolving payment flows to catch regressions before production.
- Evaluating high performance networking protocols (Homa, RDMA) with distinguished engineers in a research initiative targeting data center deployment and open source contribution.

DataServ | St. Louis, MO

DevOps Intern

May 2025 - Aug. 2025

- Wrote Python automation scripts to migrate 3,000+ SQL stored procedures across 71 client environments into Bitbucket, enabling code reviews and rollbacks for previously unversioned code.
 - Converted time-based jobs to event-driven triggers for 7 clients, contributing to a company-wide effort that cut client data turnaround from 24 hours to under 6.
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Projects

Rapid Photonic Innovation Device (RAPID) | ECE Capstone Design Award Winner | [GitHub](#)

Aug. 2025 - Apr. 2026

Converted a Blu-ray drive into a ~\$500 FPGA-controlled laser lithography system with ~0.1mm resolution

- Owned all software and supported FPGA integration on a 4-person team, spanning embedded C and Python.
 - Built bare-metal C firmware on a Zynq SoC for motor and laser control, syncing laser timing to disc rotation.
 - Designed a custom UART protocol with error checking and flow control to stream patterns from the PC.
 - Built a real-time Python GUI for live pattern visualization, manual control, and system telemetry.
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Leadership

IEEE Student Branch | *Vice President*

Aug. 2025 - May 2026

- Revived a dormant student chapter, growing membership from under 10 to 30+ through outreach and events.
- Captained the team that won 1st place at the 2026 IEEE Region 5 Cyber Challenge.

SLU Esports | *President and Teams Coordinator*

Aug. 2024 - May 2026

- Grew membership to 70+ and coordinated competitive rosters across 10+ titles.
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Skills

Programming Languages: C, C++, Python, Java, Go, SQL

Tools/Platforms: Spring, Kubernetes, Docker, MCP, Git, Linux

Hardware: VHDL, FPGA, ARM & AVR Microcontrollers

Spoken Languages: Spanish (Advanced), Korean (Intermediate)